Aman

Roll No.:220002004
B.Tech
Electrical Engineering
Indian Institute Of Technology, Indore

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Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. Major	Indian Institute of Technology Indore	6.39 (Current)	2022-Present
Senior Secondary	HBSE Board	90.2%	2021
Secondary	HBSE Board	90.8%	2019

Projects

- **Grey-box Approach for Modeling of Converter Interfaced Power Systems** *May 2025 - Present*
  Prof. Lokesh Kumar Dewangan Github
  - Grey-Box Power System Modeling: Created a validated grey-box model of a power converter by combining simulation results (Python/PSCAD) with a mathematical model (MATLAB).
  - Automated Frequency Analysis: Built a Python framework to automate PSCAD simulations, determining the converter's admittance matrix by injecting small-signal voltage perturbations and measuring the resulting current response.
  - Data Processing & Visualization: Processed simulation data using Python (NumPy, Matplotlib) and FFTs to generate frequency-domain models and Bode plots for analysis.
  - System Stability Assessment: Used the validated model to analyze power system stability with Nyquist criteria and Eigenvalue decomposition

- **Global Superstore Sales Analytics** *May - June 2025*
  MySQL, Power BI Github
  - Designed and implemented an end-to-end MySQL analytics system using Global Superstore data, applying data normalization, indexing, and window functions to compute key business KPIs.
  - Built an interactive Power BI dashboard by integrating MySQL data to visualize sales trends, profit analysis, customer performance, and regional insights using Power Query.
  - Optimized analytical reporting through star-schema modeling and SQL performance tuning, enabling faster insights into top customers, loss-making products, discount impact, and shipping efficiency.

- **Mental Wellness Web App -SoulWisdom** *Aug 2024 - Feb 2025*
  React, Tailwind CSS Github
  - Built a responsive web app providing mental wellness resources and personalized support.
  - Designed an intuitive UI with React to enhance user engagement.
  - Utilized Tailwind CSS for fast and easy styling with a responsive design.
  - Deployed the app for real-time use and feedback.

Technical Skills

- **Programming**: Java, Python, SQL(MySQL)
- **Tools/Platforms:**: PSCAD, MATLAB, Inkscape

Key courses taken

- **Electrical Engineering**: Power Electronic, Power System, Control Systems, Signal & System

Positions of Responsibility

- **Core Member**, Basketball Club, IIT Indore *Jan 2023 – Present*
- **Member**, InterIIT Sports Meet IIT Bombay'23, IIT Kanpur'24, IIT Madras'25 *2023 – 2024*

Achievements

- **Winner**, Basketball AtharvRanbhoomi'24, IIM Indore *2024*